Using a deep fusion model combining hyperspectral imaging and X-ray fluorescence to optimise the purity of recycled precious metals.

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Abstract

This paper discusses the use of hyperspectral imaging (HSI) and X-ray fluorescence (XRF) to identify high-purity metals in e-waste. It will do this by first identifying the metal samples from plastics and other less valuable materials of the circuit boards and then using XRF of these identified areas to identify the high purity metals to be separated and used for manufacture to close the supply chain. The paper will then cover the current methods the industry currently uses, including current optical separation techniques, and finally covers how xRF is worked out, and how it can be simulated to help this project. Finally, it will cover the comparison between the current techniques used in the industry and the new proposed technique.

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1 Introduction

The issue of recovering high purity precious metals is a big issue in the drive for closed supply chain loops in complex technology manufacturing. The current issue with this, is the need for very high purity precious metals, such as gold and silver, in the manufacture of new electronic devices, paired with the current inability to recycle the precious metals at high enough purities, preventing them to be reused in the manufacturing process. If the metals used in chip manufacture aren't adequately pure, it can drastically alter the electrical properties of the metal, resulting in potential device failures if used on critical parts of the chip. Some possible contaminants such as iron and nickel can diffuse rapidly into the silicon wafers at process temperatures [11], leading to defects and destroying the functionality of the chip.

As AI and machine learning technology gets better, companies are looking at better, more efficient ways of recycling the precious metals in order to create a closed loop supply chain (CSC). Manufacturing chips and other electronic components can require purities as high as 99.999% pure, which is a difficult level to achieve from recycled metals. While this may not currently be a problem, potential issues in the future, including geopolitical issues, may prevent companies from getting these ultra-pure metals. Therefore the strive to close the supply chain of these products is greater than it ever has been. These potential geopolitical issues are a big factor in obtaining precious metals as China was the largest producer in the world in 2024 and accounted for around 10 per cent of total global production of gold [4], alongside Russia the number two gold producer in the same time period. Making two of the largest gold mining countries in the world potential risks due to current and potential future conflicts.This could lead to massively raised gold and other precious metal prices, thus increasing the cost of manufacture.

As will be discussed in the literature review below, most current studies and technology only use basic RGB optical sensors, which can detect some visible differences between the parts of a PCB but lacks to differentiate anything beyond that such as metal types and qualities. This can lead to separation processes which are slow and laborious, most often by hand which takes time and human labour, potentially exposing them to harmful toxins and chemicals when undergoing the separation process.

2 Literature Review

This section of the paper will discuss what hyperspectral imaging (HSI) and X-ray fluorescence (XRF) are and how they can be used to aid the recycling of used electronics. It will then cover the current methods the industry currently uses, including current separation techniques, and then finally covers how XRF is worked out, and how it can be simulated to help this project. Finally it will cover the comparison between the current techniques used in the industry and the new proposed technique.

2.1 What are HSI and XRF and how can they be used to identify high purity metals

Hyperspectral imaging (HSI) is an advanced optical sensing technique that assembles spectroscopy and digital photography into a single system.[5] It uses both visual light and parts of the light spectrum that aren't visible which are used to identify a range of materials. The Camera captures the light intensity, radiance, for a large number of contiguous spectral bands [6]. The spectral signature of each material can be identified and makes potential similar looking materials in the visible light spectrum, much more clearly identifiable when analysed using the spectral data. The cameras capture a wide range of the light spectrum, from ultraviolet to infrared including the visible light spectrum. These values called a spectral signature, are stored in each pixel of an image which gives the opportunity for identification for the materials captured.

X-ray fluorescence (XRF) is also a technique used for identifying materials, it can either be used to qualitatively analyse materials and also quantitatively analyse materials. In a sample of material, it can identify each material in the sample, along with the quantities of each material. This can

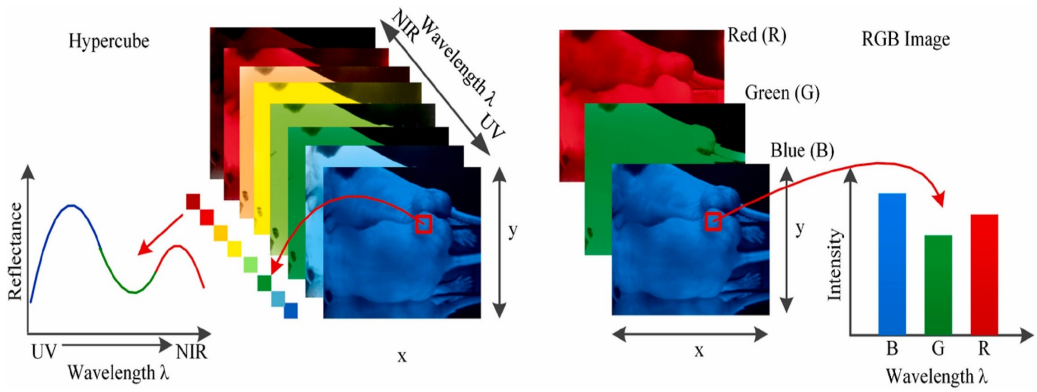

Figure 1: HSI wavelengths vs RGB.[1]

help determine the chemical composition of a sample [9] without affecting the sample, as it is a non-destructive process. It is widely used in quality and process control [9]. The X-rays eject electrons from their energy shell into lower-energy shells which can eject excess energy as a photon of X-ray light. This excess energy released is unique to the element it came from, giving the machine and the user the ability to identify the material that is being analysed.
These two technologies can be used together to aid the separation of the used electronics. HSI will be mainly used to seperate the precious metals from the epoxy and plastics. Due to the ability of HSI to primarily differentiate materials from others, it is much better suited for the identifying the different materials from the electronics. Whereas XRF can be used to quantify different metals in a sample, so it can be used for analysing the makeup of samples, allowing for purity testing of each sample of metal identified by HSI. The XRF values would ensure the high purity of the metals because of the accuracy of the process. Combined, they would be able to accurately identify high purity metals that would be able to be used for recycling and manufacturing.

2.2 How the industry currently identifies and sorts precious metals from e-waste

One method that the industry uses, which my project will build on, is the traditional use of optical sensors. Current optical sorting technology, inspects waste materials detecting changes in its appearance and its composition. This allows the sensors to detect different parts of waste in order to help aid in the separation and sorting of the waste. Using the visible light spectrum, cameras use images of the waste, to try and determine between valuable parts of the waste and the excess parts, including the fibreglass and other low-value materials. One major problem with this technology is black plastic blindness. Due to the nature of optical sensing, some black plastics can absorb all light specifically in the near-infrared range, leaving none to reflect back to the sensors.[2] This leaves the optical sensors unable to determine between different plastics.
Another method the industry currently uses is Eddy current separation. This technique is widely used later in the supply chain after e-waste is crushed. This technique uses the principle of electromagnetic induction [14]. When conductors enter the machine, they create their own magnetic field with the help of eddy currents, these magnetic fields allow for electrically conductive materials to be separated from non-metals. This generally leads to metals being separated from non-metals. This can be very beneficial as generally the more valuable components of the e-waste is the metal. However, one big issue with using the eddy current separation method is the inability to separate the metals from each other once they are separated from the non-conductive material. As it is dust form, this can pose real problems as the potential high value metals would be contaminated with other lower value metals.
Another method that is used, and perhaps the simplest is hand sorting and separation of the e-waste. This will be done by skilled workers who are very familiar with the e-waste and circuit boards, so will know from experience where the precious metals will be found. This can be beneficial as no outside technology is needed in order to have efficient separation. However this method has many big

drawbacks. As many circuit boards contain toxic chemicals, manual separation can lead to exposure to these toxic chemicals without proper PPE. [12] On top of this, without exact knowledge of where the precious metals are, just going off previous knowledge, the separation could be low quality leading to improper separation, leaving contaminants with the incorrectly separated valuable metals. This would also be very time consuming, as the intricacy of the circuit boards would be difficult for workers to work around and separate accurately.

2.3 How XRF values can be simulated

In this project, the XRF values will be simulated because of the inaccessibility of the technology as well as the lack of publicly available datasets that directly relate to this topic.
One way which XRF values can be simulated is by using the monte carlo method.[17] This works by predicting the path and interactions of individual X-ray photons, within a precisely defined sample and instrument geometry.[13] This method is used because it is most accurate for PCB analysis, as it is a complex sample, meaning other, more simple, models may struggle to factor in the complexity of the PCB models. A monte carlo model requires very detailed input parameters, mainly an accurate sample geometry and composition. For a PCB board, each sample must include the thickness of each material and the layout of what material is where. Another key parameter is the setup of the X-ray tube, this can include the voltage of the sensor, as well as the anode material. Alongside this, the physical arrangement of the beam of X-rays is key in order to accurately model the XRF values from a specific sample.
In order to accurately achieve these values, a detailed knowledge of the makeup of circuit boards is needed. The first key thing to identify is whether the PCB is single layer or multi-layer. This is key as it determines whether there is a copper layer on both sides, or just one. On multi-layer PCBs, both the top and the bottom have a copper layer, meaning it is more complex to recycle, but it contains more valuable material. The substrate of a PCB acts as the foundation upon which a printed circuit board is built[10]. The material of the substrate can have a big effect on performance as well the strength and durability of the PCB. These substrates are usually dielectric composite structures, which mainly comprise of epoxy resin. In order to consider these materials, there are many properties of PCB substrates that need to be taken into account. They can be split up into thermal properties and electrical properties.[10]

Thermal Properties	Electrical properties
Glass transition temperature (Tg)	Dielectric Constant (Er)
Decomposition Temperature (Td)	Surface Resistivity (ρS)
Coefficient of Thermal Expansion (CTE)	Electrical Strength
Thermal Conductivity (k)	

As seen in the table many factors can determine the choice of materials for the substrate of a PCB board. The Glass transition temperature (Tg) determines the temperature at which the polymer transitions from being solid to a softer state[10]. This is very important as it determines the operating temperature that the circuit board can be used at. Once a material reaches above its Tg, not only does the material lose its rigidity, but its electrical properties also decline. The second, its decomposition temperature (Td) which specifies the temperature that the polymer will start to degrade. While this is not important for the everyday use of PCBs because the Td is typically much higher than the Tg, so, while in use the circuit boards are unlikely to reach these temperatures. This however, can give an insight into the materials ability to withstand heat. The third metric, Coefficient of thermal expansion (CTE) is much more important. This determines the potential expansion the material will undergo when heated. This is a crucial metric as it can help manage reliability of the circuit boards. In order to ensure reliability, the CTE of the substrate must ideally match, or be close to, the CTE of the copper on the boards, along with any other materials on the circuit boards[16]. A failure to match these values can cause failures and stress around via holes and any solder joints. The

final thermal property, thermal conductivity highlights how well the material conducts heat. This is important when the material of a substrate is considered, as a higher thermal conductivity allow for better heat dissipation, allowing specific hotspots to be spread to neighbouring areas of the board which allows for better overall performance. In most circuit boards, a mqaterial with a lower thermal conductivity, as the heat demands of typical processors and other electrical components wouldn't require such good thermal conductivity, whereas high performance use cases may require much better thermal conductivity. Naturally, there are trade-offs between these thermal properties. Mainly due to cost, the priority of some of these properties may lead to limitations in other areas such as complexity. The electrical properties are also important when considering a substrate material. The dielectric constant (Er) determines its ability to store electrical energy and its ability to act as an electrical insulator. This can be important when determining the material based on use cases. Higher speed applications would need lower Er values, whereas typical uses may use higher value. The surface resistivity is a measure of electrical resistance of the surface of the material [7]. This can be affected by temperature and moisture. This is key as it helps ensure leakages of electric current is minimized. The next thing, and main thing, to identify is where the precious metals are typically used on a PCB board. One main use for precious metals on PCB boards is for a surface finish. This helps prevent the copper, used for connectivity, from oxidising. These precious metals are typically very unreactive, which allows for them to be used as a shield to cover the reactive copper. There are different techniques companies can use in order to surface finish which include varying materials. The first technique showed in the diagram is ENEPIG. This stands for electroless nickel electroless

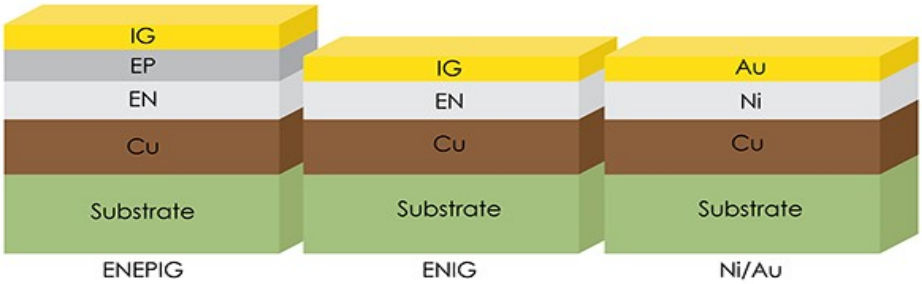

Figure 2: Makeup of materials for different surfacing techniques.[3]

Palladium immersion gold. This uses these materials in order to shield the copper. The layer of nickel acts as the main protection for the copper from oxidation. The palladium and gold then protect their corresponding under layers. The reason for the ENEPIG method is to protect from the "black pad" issue encountered in the ENIG method[8]. The ENIG as seen in the diagram, doesn't use palladium, but only electroless nickel and immersion gold. This still protects copper from oxidising and is common in circuit boards. This method opens up the possibility for the black pad issue to occur. The final method shown is electrolytic nickel/ gold. This used hard or soft gold in order to encapsulate the nickel and copper. The soft gold is very pure up to 99.9 % purity, while the hard gold is a gold alloy, which is used for heavy usage areas[3].

Combining this knowledge, the XRF values can be accurately simulated to aid the identification of pure precious metals. However, in order to accurately model for the simulations, it is important to understand how XRF works. As mentioned previously X-rays are used to analyse different materials, with a unique signature, and can identify materials in a sample. This is based on the excitation of atoms in the sample [15]. With many atoms being present, an array of X-rays with varying energies are emitted, these create signals that are collected using a multi-channel analyser [15]. The signals are converted into a spectrum, where the intensity of peaks in signals received can identify elements in the sample. The XRF detectors rely on a good sample rate, as they analyse each X-ray one by

one, this can massively slow down the process. To tackle this X-ray counts can be adjusted in order to counteract slower detection speeds, however this would come at the expense of accuracy in the testing. As seen in the table, the relative error percentage has a direct correlation with the number of

Number of counts N	Abs. error √N	Rel. error in %
100	10	10 %
10,000	100	1 %
1,000,000	1,000	0.1 %

Figure 3: Table of errors vs X-ray count.[15]

X-ray counts. The accuracy and timely calculation of results rely on a sensor with a high-count rate, which may also affect the time taken to simulate the XRF values. While the values won't actually be detected, it may affect the accuracy of the simulations.

2.4 Conclusion

Table 1: Comparison between proposed method and current methods

Method	HSI + XRF	Electrostatic Separation (ES)	Manual Dismantling & Sorting	Eddy Current Separation (ECS)
How it separates precious metals	Isolate and verify specific, high-value components to create a high purity, small-volume concentrate before refining.	Separate all metal powder including both precious and non-precious metals from plastic/fiberglass fines.	Physically remove high precious metal components prior to shredding.	Recover ferrous metal from pre-shredded e-waste, leaving precious metal and non-ferrous metals
Precious Metal Purity of Output	Highest Achievable Pre-Refining Purity: XRF verifies individual fragment/component content, allowing sorting to extremely tight specifications.	Moderate: The output is a mixed metal powder so less pure precious metals	High: Dependent on the skill of the sorter; the precious metal is still part of the original component structure.	Low : precious metal is mostly non-magnetic and non-bulky, so it often remains in the fine non-metallic stream
Mechanism of Separation	Material Signature (HSI) + Elemental Composition (XRF)	Electrical Conductivity	Visual Recognition	Induced Eddy Current, magnetic force on ferrous objects
Loss Potential of precious metal	Very Low: HSI identifies precious metal-rich components, allowing precise physical separation before destructive processes	Moderate: Some precious metal-containing fines can be lost in the non-conductive (plastic/fiberglass) fraction	Zero: If the component is identified and removed intact, loss is prevented	Low: precious metals are typically found on fine particles and will
Speed/Throughput	High: HSI is a continuous, high-speed sorting system.	Moderate: Requires highly specialized, fine, dry feedstock.	Slow/Labour-Intensive: Difficult to manage a high throughput of items	Very High: Excellent for bulk, continuous processing of coarser fragments.
Best Use Case	Refining Feedstock Optimization: Creating distinct, high-value batches from a mixed metal concentrate.	Fine Grinding: Ensuring no small precious metal fragments are lost in the non-metal stream post-shredding.	High-Value Items: Isolating CPUs, memory sticks, and boards before mechanical processing.	Base Metal Removal: Clearing out large quantities of low-value Aluminium/Copper to concentrate the high-precious metal fines for further processing.

To conclude, there is a clear gap in knowledge regarding the combination of sensing methods to produce high quality results from the recycling of circuit boards. Existing methods, as discussed previously, have led to the inaccurate identification of high purity precious metals. Through the inability to accurately identify both where the precious metals are on the recycled electronics, and the purity of these metals, the ability to reuse these metals in the closed supply chain is weakened. Currently metals cannot be recovered at high enough purities to be reused in the manufacture of high-quality electronics such as processors and microchips. The metals currently can only be used as less important parts of less specialised use cases, which do not require such high-quality metals.
To close the gap in this research, my project proposes to fuse both hyperspectral imaging and XRF testing to identify high purity metals in samples of circuit boards. It will do this by first identifying the metal samples from plastics and other less valuable materials of the circuit boards and then using XRF of these identified areas to identify the high purity metals to be separated and used for manufacture to close the supply chain.

3 Project Specification

The problem that this project aims to solve is only just becoming an issue in the present and near future. In the past manufacturers have had less pressure on trying to create a closed supply chain (CSC). This is due to the lack of rising geopolitical pressures alongside less emphasis on protecting the environment. The push to create these CSCs is in order to help stabilise the supply of essential production of high-level chip manufacturing, which may be put at risk due to these factors. In more recent history the push for recycling of e-waste has risen, however the recycling methods rely on extracting the precious metals in any form, rather than the high purity that my method will deliver. The motivation behind this project is to move the industry closer to being a CSC. As mentioned before, the project aims to help reduce the impact of external factors in the supply chain of high-level microchips. As the need for extreme levels of purity in the manufacture, this project aims to be able to help recover these high purity metals from e-waste. It will do this by combining hyperspectral imaging and X-ray fluorescent scanning in order to identify high-purity metals in e-waste. Which can then be separated and reused in high-level manufacturing.

3.1 Aims and Objectives

- Produce XRF simulated values for potential samples provided from circuit boards
- Produce a combined deep learning model to analyse PCB boards to identify high-purity metals
- Evaluate the findings using existing knowledge

3.2 Methodology

First HSI data will be collected from datasets readily available online. This will be for the HSI part of the fusion model. This will be checked for compatibility with the task of separating metals from the extra PCB material, making sure that the spectral range is suitable. This data will need to be the raw reflectance values, not the digital numbers. The spatial consistency also needs to be checked, making sure the resolution is consistent across the dataset. The HSI data must then be extracted to be processed by the fusion model. Small, fixed sized patches of the data are then used as the individual inputs for the CNN. In order to prevent overfitting, some level of data augmentation will be used to help generate some difference in then datasets used. These patches of data must then be labelled with their sorting category.

Next the XRF must be simulated. As the HSI data is from an existing source, and the XRF values are to be simulated. Using the monte carlo simulation method, true XRF values need to be simulated. For each HSI image, composition mapping will need to take place, identifying specific components of the circuit board. The composition of these parts will then need to be defined, based on previous knowledge of these components and their elemental makeup. The monte carlo simulations are then run, based on these compositions, returning the composition of a sample. The synthetic XRF values then align with the HSI inputs for the fusion network.

The feature extraction stage of the process will be run in parallel. The two inputs, HSI and XRF, will be fed into separate branches of the deep learning model. The HSI branch will be made up of a 3D convolution neural network (CNN). This will be used to extract the spatial features of the circuit boards. It will mainly be used to identify the geometry of the chips as well as potential contaminants. The layers of this network would include a convolutional layer, which turns the image into data. Following this, a normalisation layer, which makes the outputs of the previous layer normalised. There would then a ReLU layer, another activation layer, which helps the network to compute more complex patterns. A max pooling layer is then included in order to reduce the computational burden by reducing the spatial size of the data. Another 3D convolutional layer follows the pooling layer; this is used to produce a compact representation of the output. A Flattening layer finishes the CNN, transforming the 3D data into a one-dimensional vector, which acts as the output, capturing the

material matrix and a contaminant risk.
For the XRF branch of the fusion network, a fully connected network will be used. Its function will be to learn the elemental ratios, in order to confirm high-value concentrations of these metals. The fully connected network will consist of a fully connected layer, used to analyse the input vector. Followed by a normalisation layer and a ReLU layer in order to normalise the data and help identify more complex patterns. Finally, another fully connected layer is used to close this branch of the fusion network.
To complete the fusion network, the data from the two branches must be combined. The fusion layers, consisting of fully connected layers, combine the data to create a final correlation. The final layer, the output layer will use a SoftMax activation function to distribute a probability between multiple sorting categories. A high-value, high-purity stream, a base metal stream. A stream which contains toxic contaminants, and finally a waste stream for plastics and other low value materials.
This multi-streamed output allows for good separation of the different materials, which can help enhance the supply chain further when the extraction of the material happens. The high-purity stream will allow for high purity precious metals be separated from the other metals, potentially allowing them to be reused again in high-tech manufacturing.

3.3 Schedule and Milestones

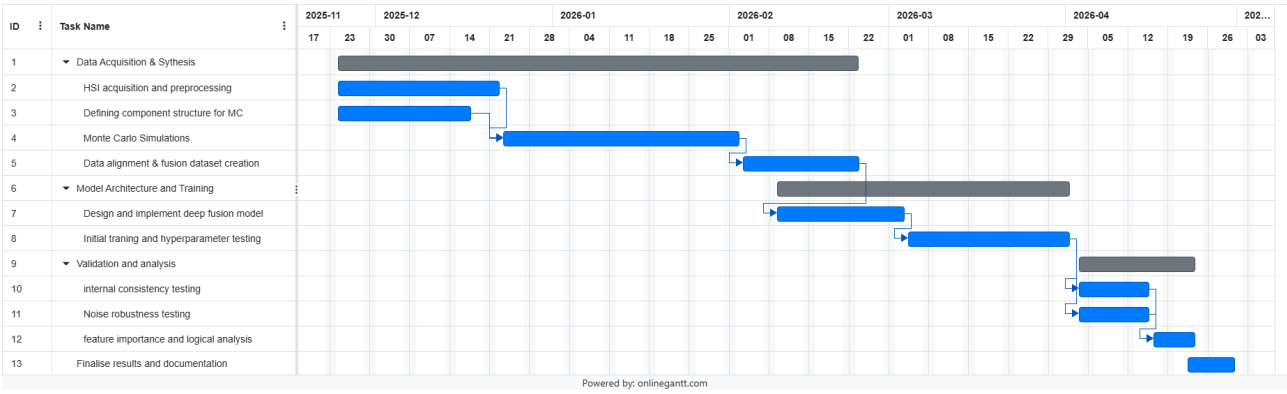

Figure 4: Gantt chart for the project

As seen in this Gantt chart, this project comprises of many steps. The arrows between tasks represent the dependencies. The steps are as follows:

1. Obtaining and preparing the data for the model
   This includes the preparation of the HSI datasets as well as setting up the simulations. The simulations depend on the definition of the component structure. The main milestones of this step are correctly pre-processed HSI values along with accurately simulated XRF values

2. Creating the Model
   Designing and implementing the model requires the data to be created and prepared for it. The milestone for this step is the model being created and producing accurate results

3. Validation and Analysis
   This is where the model is evaluated and tested. The milestone for this section, is good analysis of the model and the required results.

3.4 Validity and reliability

In order to test the reliability of the model, changes in the data and features of the data given can be used. By changing the consistency and quality of the data, the model's reliability can be tested. Adding noise to some spectral bands for the HSI data, as well as some XRF noise as well, the model

can then be tested to see if it is still reliable given imperfect data. The model should maintain a high classification consistency even with the changed inputs, which will then confirm the reliability of the model.
To test the reliability of the monte carlo simulations, multiple simulations of the same structures can be run, but with varying angles of the sensor or varying thicknesses of the material. The goal for this testing is to ensure the XRF values aren't overfit to this specific geometry and are accurate to real world tests.
Validity testing of the model is also very important. Ensuring that the model can perform well on unseen sample is key. By splitting the dataset up and training the model multiple times for different parts of the dataset, it can ensure the model is actually learning from the data. Another way to test the validity is by pushing the limits of the technologies used. XRF sensing can struggle if sample sizes are too small, so by giving it small samples to try and analyse can give an insight into the effectiveness of the model.
The model's success will be measured on the ability to identify high purity metals. Given the previous knowledge on the typical levels of high-purity metals, it can be used to identify how much of the perceived high-purity metal is extracted based on how much was put into the sample. The output streams of samples will be checked with percentages of precious metals in the circuit boards, and how much is included in the output streams.

3.5 Ethics and Risk Management

This project only includes readily available datasets of HSI information so there are no inherent ethical issues. The datasets that will be used under the permission of the owner. Due to the theoretical nature of the project, there will be no physical risks to manage. The only risks that may be an issue is the lack of the size and availability of suitable datasets. Due to the obscurity of the problem, there is a small number of datasets available to use. Another issue comes with the simulation of the XRF values, the simulations may be inaccurate due to lack of knowledge of the area. Lots of research will be done to try and tackle this, but due to the nature of a pre-existing dataset and simulated values, the simulations may not be 100% correct.

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